Business Question and Visualization Report

Date	28 October 2026
Team ID	[Insert Your Actual Team ID, e.g., TM-1042]
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

1. **What is the distribution of patients across Age Brackets and WHO BMI categories?**

   - *Code*

```
df = pd.read_csv('smart_healthcare_dataset.csv')
df['Age_Bracket'] = pd.cut(df['age'], bins=[17, 34, 50, 65, 100],
                           labels=['18-34', '35-50', '51-65', '65+'])
df['BMI_Category'] = pd.cut(df['bmi'], bins=[0, 18.5, 25, 30, 100], right=False,
                            labels=['Underweight', 'Normal Weight',
                                    'Overweight', 'Obese'])
pd.crosstab(df['BMI_Category'], df['Age_Bracket'])
```

- *Screenshot of visualisation*

02Age_BMI_Distribution

BMI Category	18–34 (Young Ad..	35–50 (Middle A..	51–65 (Senior)	65+ (Elderly)
	Age Brackets			
Normal Weight	552	524	485	449
Obese	202	202	195	204
Overweight	436	448	398	389
Underweight	134	127	133	122

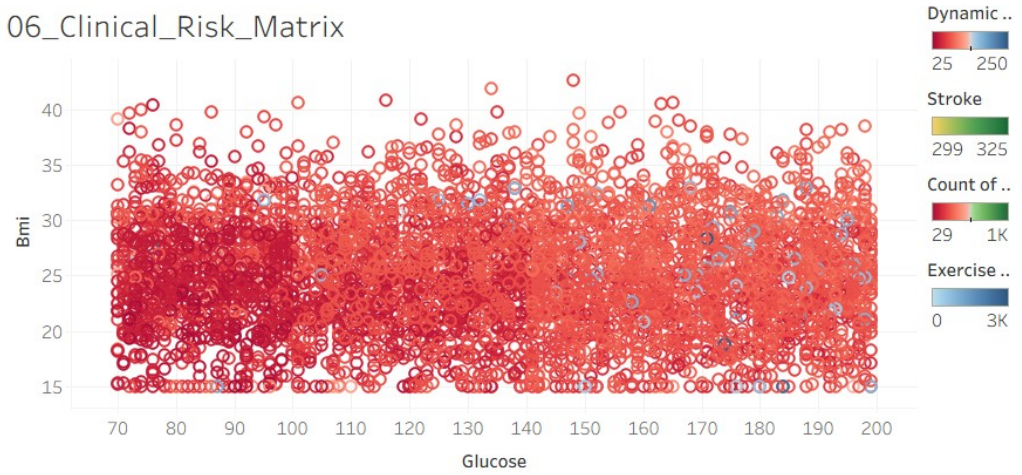
Visual Finding: Normal Weight is the largest BMI group in every age bracket (552, 524, 485 and 449 patients for 18–34, 35–50, 51–65 and 65+), followed by Overweight; Obese patients stay steady at about 195–204 per bracket. The cohort is split fairly evenly by age, from 1,324 patients aged 18–34 to 1,164 aged 65+.

2. **How do clinical vitals (Blood Pressure & Glucose) correlate with the Dynamic Risk Tier Index?**

- *Code*

```
def assign_risk(row):
    if row['blood_pressure'] >= 140 or row['glucose'] >= 140:
        return 'High Risk'
    elif row['blood_pressure'] >= 120 or row['glucose'] >= 100:
        return 'Moderate Risk'
    return 'Low Risk'
df['Risk_Tier'] = df.apply(assign_risk, axis=1)
df.groupby('Risk_Tier')[['blood_pressure', 'glucose']].mean().round(1)
df[['blood_pressure', 'glucose']].corr()
```

- *Screenshot of visualisation*



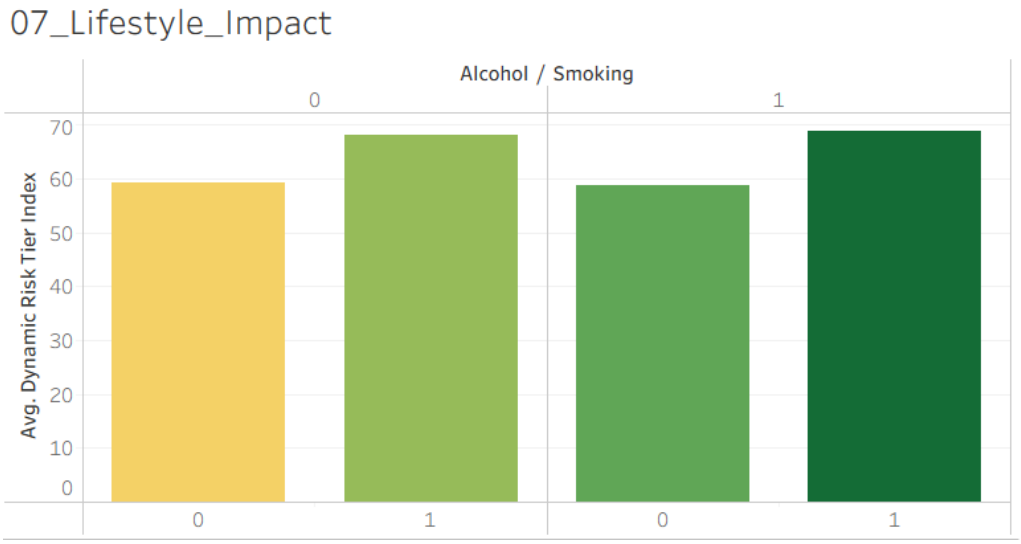
Visual Finding: High Risk patients sit at or above the 140 mmHg blood pressure or 140 mg/dL glucose thresholds on the scatter plot matrix. Average vitals step up with tier: blood pressure 104.7, 117.6 and 143.3 mmHg and glucose 85.0, 111.9 and 147.5 mg/dL for Low, Moderate and High Risk. Blood pressure and glucose are almost uncorrelated with each other (r = 0.02), so each independently drives the tier.

3. **What is the impact of lifestyle habits (Smoking & Alcohol) on chronic disease risk?**

- *Code*

```
lifestyle_risk = (df.groupby(['smoking', 'alcohol'])['Risk_Tier']
                    .value_counts(normalize=True)
                    .unstack()
                    .round(3))
```

- *Screenshot of visualisation*



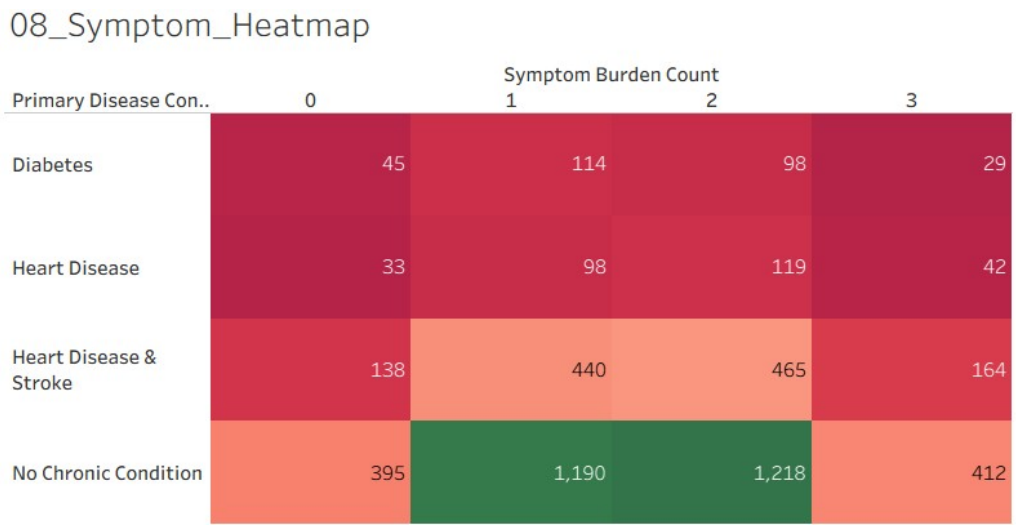
Visual Finding: Under the blood pressure and glucose rule, the High Risk share is similar in all four smoking and alcohol groups (67.7% to 70.3%), so there is no strong lifestyle effect on the tier itself. The lifestyle dashboard, however, shows a higher average Dynamic Risk Tier Index for smokers (roughly 68–70 versus about 59 for non-smokers), so the index captures a smoking effect that the threshold rule does not.

4. **What is the co-occurrence pattern of multi-condition symptom burdens across cohorts?**

- *Code*

```
condition_cols = ['heart_disease', 'diabetes', 'stroke']
symptom_cols   = ['fatigue', 'chest_pain', 'dizziness']
condition_corr = df[condition_cols].corr().round(2)
df['Symptom_Burden'] = df[symptom_cols].sum(axis=1)
pd.crosstab(df['Symptom_Burden'], df['Risk_Tier'])
```

- *Screenshot of visualisation*



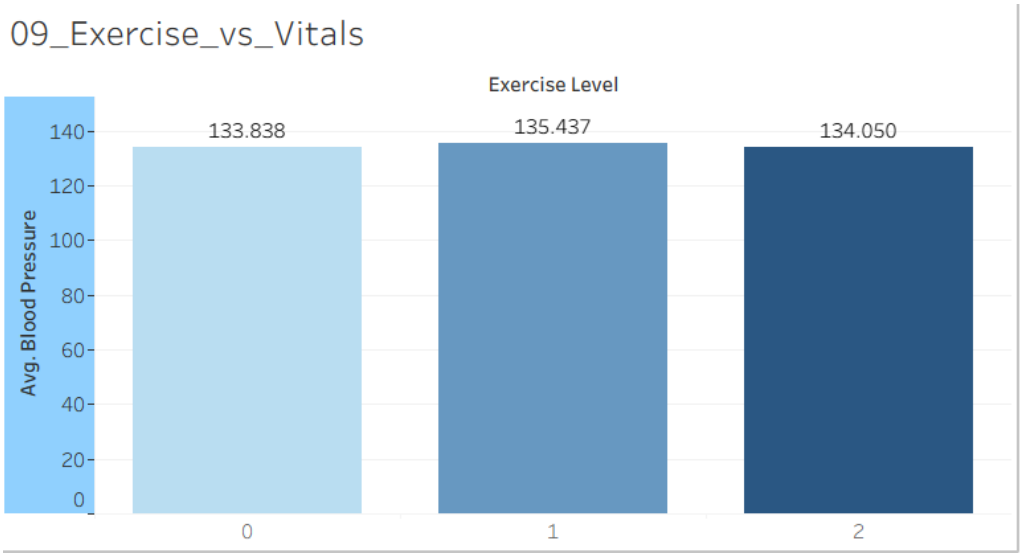
Visual Finding: The three chronic conditions are strongly correlated with one another (heart disease and diabetes r = 0.86, heart disease and stroke r = 0.84, diabetes and stroke r = 0.79), whereas the symptoms (fatigue, chest pain, dizziness) are essentially independent of each other and of the conditions. In the Symptom Burden Heatmap, Heart Disease & Stroke patients are concentrated at 1–2 concurrent symptoms (440 and 465 patients).

5. **How does exercise frequency influence Systolic Blood Pressure levels?**

- *Code*

```
bp_exercise = (df.groupby('exercise_level')['blood_pressure']
                 .mean().round(3).reset_index())
```

- *Screenshot of visualisation*



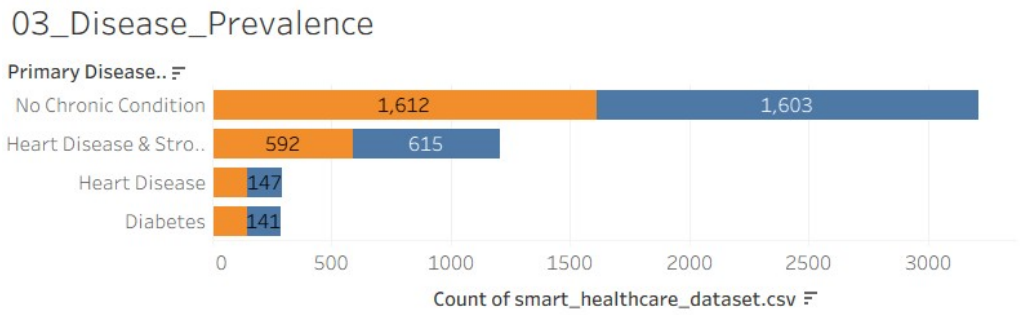
Visual Finding: Average blood pressure is almost flat across the three exercise levels (133.838, 135.437 and 134.050 mmHg for levels 0, 1 and 2), a spread of only 1.6 mmHg. This dataset therefore shows no meaningful blood pressure benefit from higher exercise level.

6. **What is the overall disease prevalence rate across age demographic brackets?**

- *Code*

```
disease_prevalence = (df.groupby('Age_Bracket', observed=True)
                        [['heart_disease', 'diabetes', 'stroke']]
                        .mean() * 100).round(1)
```

- *Screenshot of visualisation*



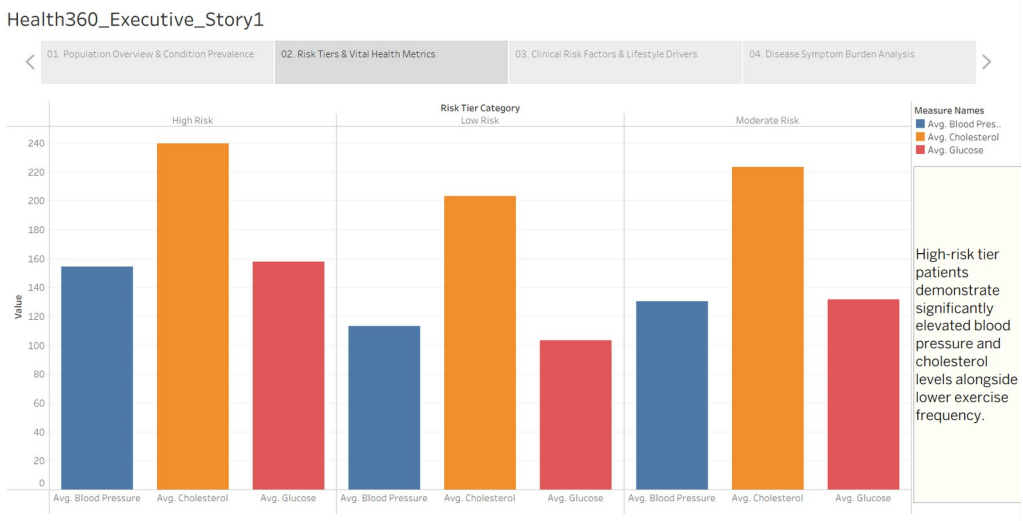
Visual Finding: Prevalence rises sharply with age. No patient aged 18–50 has a recorded condition, in the 51–65 bracket diabetes is 48.1%, heart disease 29.6% and stroke 6.5%, and in the 65+ bracket prevalence is 97.9% for heart disease and 100% for diabetes and stroke. Overall, 3,215 patients (64.3%) have no chronic condition.

7. **How are hospital resources triaged based on patient risk stratification?**

- *Code*

```
triage_summary = df['Risk_Tier'].value_counts(normalize=True) * 100
triage_summary.round(1)
```

- *Screenshot of visualisation*



Visual Finding: The cohort splits into 69.6% High Risk (3,479 patients), 22.5% Moderate Risk (1,123) and 8.0% Low Risk (398), which sets the baseline for resource allocation. High Risk patients also show the highest average blood pressure, cholesterol and glucose in the Risk Tier scene of the Executive Story.

8. **How does the 4-scene Executive Story guide leadership in decision-making?**

- *Code*

```
function navigateToStoryPoint(sceneIndex) {
    workbook.getActiveSheet().getStoryPointsInfo()[sceneIndex].activateAsync();
}
```

- *Screenshot of visualisation*



Visual Finding: The story moves leadership through four scenes: 01 Population Overview & Condition Prevalence, 02 Risk Tiers & Vital Health Metrics, 03 Clinical Risk Factors & Lifestyle Drivers and 04 Disease Symptom Burden Analysis. It runs from a population-level view down to the co-occurring conditions that need clinical attention.

Note: Min 8 business question and visualisations Required inform of above